

$$-hA_s(T-T_f) = \rho V C_p \frac{dT}{dt} \quad T(t)$$

$$t=0 \quad T(0) = T_{ini}$$

Lumped capacitance.

$$\int_{T_{ini}}^{T(t)} \frac{dT}{T-T_f} = \int_0^t \frac{-hA_s}{\rho V C_p} dt$$

$$\ln \left(\frac{T-T_f}{T_{ini}-T_f} \right) = - \frac{hA_s}{\rho V C_p} t$$

$$\frac{T-T_f}{T_{ini}-T_f} = e^{-\frac{hA_s}{\rho V C_p} t}$$

$$\frac{hA_s}{\rho V C_p} [s] = \frac{\frac{10^5}{m^2 K} \cdot m^2}{\frac{10^3}{m^3} \cdot m^3 \cdot \frac{J}{kg K}} = \frac{1}{s}$$

$$\theta = e^{-Bi \cdot Fo}$$

$$Bi = \frac{h L_c}{k} \quad L_c = \frac{V}{A_s}$$

$$\theta = \frac{T-T_f}{T_{ini}-T_f}$$

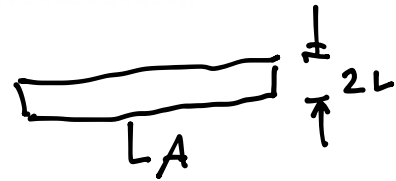
$$Fo = \frac{\alpha t}{L_c^2} \quad \alpha = \frac{k}{\rho C_p} \propto \frac{10^{-1}}{10^3 \cdot 10^3} \propto 10^{-7}$$

$$Bi = \frac{R_{cond}}{R_{conv}} \propto \frac{\frac{L}{k A_s}}{\frac{1}{h A}} \propto \frac{h}{k} \quad Bi < 0.1$$

$$\eta = \frac{\Delta T}{R}$$

5.10 A flaked cereal is of thickness $2L = 1.2$ mm. The density, specific heat, and thermal conductivity of the flake are $\rho = 700 \text{ kg/m}^3$, $c_p = 2400 \text{ J/kg} \cdot \text{K}$, and $k = 0.34 \text{ W/m} \cdot \text{K}$, respectively. The product is to be baked by increasing its temperature from $T_i = 20^\circ\text{C}$ to $T_f = 220^\circ\text{C}$ in a convection oven, through which the product is carried on a conveyor. If the oven is $L_o = 3$ m long and the convection heat transfer coefficient at the product surface and oven air temperature are $h = 55 \text{ W/m}^2 \cdot \text{K}$ and $T_\infty = 300^\circ\text{C}$, respectively, determine the required conveyor velocity, V . An engineer suggests that if the flake thickness is reduced to $2L = 1.0$ mm the conveyor velocity can be increased, resulting in higher productivity. Determine the required conveyor velocity for the thinner flake.

$$L_c = \frac{V}{A_s} = \frac{A(2L)}{A} = 2L$$



$$\frac{T_f \overset{\sim 220}{\leftarrow} T_\infty \overset{\sim 300}{\leftarrow}}{T_i \overset{\sim 20}{\leftarrow} T_\infty \overset{\sim 300}{\leftarrow}} = e^{-Bi \cdot Fo} \quad ?$$

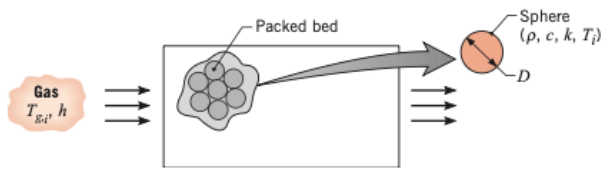
$$Bi = \frac{h L_c}{k}$$

$$\begin{aligned} \frac{dN}{dt} &= -k N \\ &= -k(T) N \end{aligned}$$

$$\frac{dw}{N} = -k(T) dt$$

$$\left(\frac{N}{N_i} \right) = e^{-k(T)t}$$

5.12 Thermal energy storage systems commonly involve a *packed bed* of solid spheres, through which a hot gas flows if the system is being charged, or a cold gas if it is being discharged. In a charging process, heat transfer from the hot gas increases thermal energy stored within the colder spheres; during discharge, the stored energy decreases as heat is transferred from the warmer spheres to the cooler gas.



Consider a packed bed of 75-mm-diameter aluminum spheres ($\rho = 2700 \text{ kg/m}^3$, $c = 950 \text{ J/kg} \cdot \text{K}$, $k = 240 \text{ W/m} \cdot \text{K}$) and a charging process for which gas enters the storage unit at a temperature of $T_{g,i} = 300^\circ\text{C}$. If the initial temperature of the spheres is $T_i = 25^\circ\text{C}$ and the convection coefficient is $h = 75 \text{ W/m}^2 \cdot \text{K}$, how long does it take a sphere near the inlet of the system to accumulate 90% of the maximum possible thermal energy? What is the corresponding temperature at the center of the sphere? Is there any advantage to using copper instead of aluminum?

$$Q_{\max} = m C_p (T_{eq} - T_i)$$

$$\frac{T - T_\infty}{T_i - T_\infty} = e^{-\frac{h A_s}{\rho C_p V} t}$$

$$\dot{q}_{\text{conv}} = h A_s (T(t) - T_\infty)$$

$$Q(t) = \int_0^t h A_s (T(t) - T_\infty) dt$$

$$Q(t) = \int_0^t h A_s (T_i - T_\infty) e^{-\frac{h A_s}{\rho C_p V} t} dt$$

$$Q = -\frac{\rho C_p V}{h A_s} h A_s (T_i - T_\infty) \left[e^{-\frac{h A_s}{\rho C_p V} t} \right]_0^t$$

$$Q(t) = -\rho V C_p (T_i - T_\infty) \left[e^{-\frac{h A_s}{\rho C_p V} t} - 1 \right]$$

$$t \rightarrow \infty$$

$$Q(t \rightarrow \infty) = \rho V C_p (T_i - T_\infty)$$

$$0.9 \rho V C_p (T_\infty - T_i) = -\rho V C_p (T_i - T_\infty) \left[e^{-\frac{h A_s t}{\rho V C_p}} - 1 \right]$$

$$0.9 = 1 - e^{-\frac{hA_s t}{\rho V c_p}}$$

$$0.1 = e^{-\frac{hA_s t}{\rho V c_p}}$$

$$\ln 0.1 = -\frac{hA_s t}{\rho V c_p}$$

$$t = -\frac{\rho V c_p}{hA_s} \ln 0.1$$

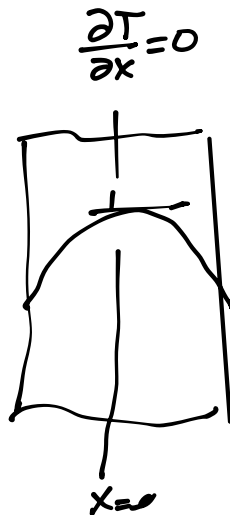
$$Bi < 0.1$$

$$\frac{h l_c}{K} < 0.1$$

$$l_c = \frac{V}{A_b} = \frac{\frac{4}{3}\pi r^3}{4\pi r^2}$$

$$l_c = \frac{1}{3}r$$

$$\alpha \frac{\partial^2 T}{\partial x^2} = \frac{\partial T}{\partial t}$$



h, T_∞

$t=0 \quad T(x,0)=T_i$

$$Fo = \frac{\overset{10^{-4}}{\alpha} t}{\underset{\downarrow}{l_c^2}}$$

$$\alpha = \frac{K}{\rho c_p} = \frac{10^2}{10^3 \times 10^3} = 10^{-4}$$

$$0.2 = 10^{-4} t$$

$$t \sim 0.2 \times 10^4 = 2000 \text{ } \underline{\underline{\text{secs}}}$$